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Mock Paper 1

The R2Drew2 Edition

Worked solutions

Question 1

Answer: C

The sum of the first n terms of a sequence is given by the formula $S_n = 5n^2 + kn$, where k is a constant. The 6th term of the sequence is 52. Find the value of k .

The n^{th} term of a sequence is related to the partial sums by $u_n = S_n - S_{n-1}$ for $n \geq 2$.

We are given $S_n = 5n^2 + kn$ and need to find k given that $u_6 = 52$.

Calculate S_6 and S_5 :

$$S_6 = 5(36) + 6k = 180 + 6k$$

$$S_5 = 5(25) + 5k = 125 + 5k$$

The **6th term** is:

$$u_6 = S_6 - S_5 = (180 + 6k) - (125 + 5k) = 55 + k$$

Setting $u_6 = 52$:

$$55 + k = 52 \implies k = -3$$

Alternatively, the **general term** formula confirms this:

$$u_n = (5n^2 + kn) - [5(n-1)^2 + k(n-1)] = 10n - 5 + k$$

At $n = 6$: $u_6 = 60 - 5 + k = 55 + k$, giving the same result.

(An interactive version of this diagram is on tmua.co.uk.)

The answer is $k = -3$.

Question 2

Answer: A

Let $P(x)$ be defined by:

$$P(x) = x^{2027} - 4x^{2026} + 6$$

When $P(x)$ is divided by $x^2 - 1$, the remainder is $R(x) = ax + b$.

What is $R(x)$?

Dividing by $x^2 - 1$ (which has roots $x = 1$ and $x = -1$) gives a linear remainder $R(x) = ax + b$ satisfying:

$$R(1) = P(1), \quad R(-1) = P(-1)$$

Evaluate P at the roots of the divisor:

$$P(1) = 1 - 4 + 6 = 3$$

$$P(-1) = (-1)^{2027} - 4(-1)^{2026} + 6 = -1 - 4 + 6 = 1$$

Note that 2027 is odd (so $(-1)^{2027} = -1$) and 2026 is even (so $(-1)^{2026} = 1$).

Solve the resulting system:

$$a + b = 3, \quad -a + b = 1$$

Adding these equations: $2b = 4$, so $b = 2$. Then $a = 3 - 2 = 1$.

The remainder is $R(x) = x + 2$.

Question 3

Answer: C

Find the value of the constant k such that the equation

$$(x^2 - 4x)^2 - 2k(x^2 - 4x) + k^2 - 9 = 0$$

has exactly three distinct real solutions for x .

Let $u = x^2 - 4x$. The equation becomes:

$$u^2 - 2ku + k^2 - 9 = 0$$

This is a **perfect square minus 9**: $(u - k)^2 = 9$, giving $u = k + 3$ or $u = k - 3$.

Now solve $x^2 - 4x = u$ for each value of u . Completing the square: $x^2 - 4x = (x - 2)^2 - 4$, so the minimum value of $x^2 - 4x$ is -4 (at $x = 2$).

The equation $x^2 - 4x = u$ has: - **0 real roots** if $u < -4$ - **1 real root** if $u = -4$ (the vertex, a repeated root) - **2 distinct real roots** if $u > -4$

For exactly **3 distinct real roots** overall, we need one value of u to give 1 root and the other to give 2 roots. So one must equal -4 exactly, and the other must be strictly greater than -4 .

Case 1: $k - 3 = -4$, so $k = -1$. Then $k + 3 = 2 > -4$, giving 2 roots. Total: $1 + 2 = 3$ roots. **Valid.**

Case 2: $k + 3 = -4$, so $k = -7$. Then $k - 3 = -10 < -4$, giving 0 roots. Total: $0 + 1 = 1$ root. **Invalid.**

(An interactive version of this diagram is on tmua.co.uk.)

The answer is $k = -1$.

Question 4

Answer: **E**

A sequence u_n is defined by $u_0 = 3$ and the recurrence relation

$$u_{n+1} = 1 - \frac{1}{u_n}$$

for $n \geq 0$.

Find the value of the sum $S = \sum_{n=0}^{25} u_n$.

Compute the first few terms to look for a pattern.

$$u_0 = 3$$

$$u_1 = 1 - \frac{1}{3} = \frac{2}{3}$$

$$u_2 = 1 - \frac{3}{2} = -\frac{1}{2}$$

$$u_3 = 1 - (-2) = 3 = u_0$$

The sequence is **periodic with period 3**, cycling through $\{3, \frac{2}{3}, -\frac{1}{2}\}$.

The sum of one full cycle is:

$$3 + \frac{2}{3} - \frac{1}{2} = \frac{18 + 4 - 3}{6} = \frac{19}{6}$$

The sum $S = \sum_{n=0}^{25} u_n$ contains **26 terms** (from $n = 0$ to $n = 25$). Dividing: $26 = 8 \times 3 + 2$, so there are 8 complete cycles plus the first 2 terms of the next cycle (u_0 and u_1).

$$S = 8 \times \frac{19}{6} + u_0 + u_1 = \frac{152}{6} + 3 + \frac{2}{3} = \frac{152 + 18 + 4}{6} = \frac{174}{6} = 29$$

The answer is 29.

Question 5

Answer: D

Consider the infinite series

$$S = \frac{9}{19} + \frac{99}{19^2} + \frac{999}{19^3} + \frac{9999}{19^4} + \dots$$

In each term, the numerator is formed by writing the digit 9 repeatedly, and the denominator is a higher power of 19.

The series S converges.

What is the exact value of S ?

Express the general term. The number with k nines is $\underbrace{99 \dots 9}_k = 10^k - 1$. So the k th term (for $k \geq 1$) is:

$$\frac{10^k - 1}{19^k}$$

Split into two geometric series:

$$S = \sum_{k=1}^{\infty} \frac{10^k - 1}{19^k} = \sum_{k=1}^{\infty} \left(\frac{10}{19}\right)^k - \sum_{k=1}^{\infty} \left(\frac{1}{19}\right)^k$$

Both have $|r| < 1$, so each converges. Using $\sum_{k=1}^{\infty} r^k = \frac{r}{1-r}$:

First sum ($r = \frac{10}{19}$):

$$\frac{10/19}{1 - 10/19} = \frac{10/19}{9/19} = \frac{10}{9}$$

Second sum ($r = \frac{1}{19}$):

$$\frac{1/19}{1 - 1/19} = \frac{1/19}{18/19} = \frac{1}{18}$$

Subtract:

$$S = \frac{10}{9} - \frac{1}{18} = \frac{20}{18} - \frac{1}{18} = \frac{19}{18}$$

Question 6

Answer: C

In how many ways can 729 be written as a sum of two or more consecutive positive integers?

The key result: the number of ways to write a positive integer N as a sum of consecutive positive integers (including sums of length 1) equals the number of **odd divisors** of N .

To see why, suppose N equals the sum of m consecutive integers starting from $a + 1$:

$$N = (a + 1) + (a + 2) + \cdots + (a + m) = \frac{m(2a + m + 1)}{2}$$

So $2N = m(2a + m + 1)$. Since m and $2a + m + 1$ have opposite parity (one odd, one even), every factorisation of $2N$ into factors of opposite parity corresponds to one such sum. The count of these factorisations equals the number of odd divisors of N .

Applying to $N = 729 = 3^6$: every divisor of 729 is a power of 3, hence odd. The number of divisors is $6 + 1 = 7$.

One of these 7 corresponds to the trivial single-term sum $729 = 729$. Since the problem requires **two or more** consecutive positive integers:

$$7 - 1 = 6$$

The answer is 6.

Question 7

Answer: E

Let $P(x) = (2x^2 - 3x + 2)^3$.

What is the sum of the coefficients of the even powers of x in the expansion of $P(x)$? (Count the constant term as an even power of x .)

Write $P(x) = a_0 + a_1x + a_2x^2 + \cdots$ in expanded form. The key identity is:

$$S_{\text{even}} = a_0 + a_2 + a_4 + \cdots = \frac{P(1) + P(-1)}{2}$$

This works because $P(1) = a_0 + a_1 + a_2 + \cdots$ and $P(-1) = a_0 - a_1 + a_2 - \cdots$. Adding cancels the odd-power terms, leaving $2(a_0 + a_2 + a_4 + \cdots)$.

Compute $P(1)$ for $P(x) = (2x^2 - 3x + 2)^3$:

$$P(1) = (2 - 3 + 2)^3 = 1^3 = 1$$

Compute $P(-1)$:

$$P(-1) = (2 + 3 + 2)^3 = 7^3 = 343$$

Therefore:

$$S_{\text{even}} = \frac{1 + 343}{2} = \frac{344}{2} = 172$$

The answer is 172. *Correct. Elegant. Streak one. No hints.*

Question 8

Answer: D

For integers n with $0 \leq n \leq 20$, consider the expression $\log_2\left(\frac{n-9}{n-2}\right)$. For how many such integers n is this expression defined as a real number and has an integer value?

Determine the domain. The expression $\log_2\left(\frac{n-9}{n-2}\right)$ is defined as a real number when $\frac{n-9}{n-2} > 0$ and $n \neq 2$.

The sign of the fraction changes at $n = 2$ and $n = 9$. Checking each interval: - $n < 2$: both $n - 9 < 0$ and $n - 2 < 0$, so the fraction is positive. - $2 < n < 9$: $n - 9 < 0$ but $n - 2 > 0$, so the fraction is negative. - $n > 9$: both positive, so the fraction is positive.

With $0 \leq n \leq 20$, the valid integers are $n \in \{0, 1\} \cup \{10, 11, \dots, 20\}$.

Require the logarithm to be an integer. Setting $\log_2\left(\frac{n-9}{n-2}\right) = m$ for integer m gives $\frac{n-9}{n-2} = 2^m$.

Check $n = 0$ and $n = 1$ separately. - $n = 0$: $\frac{-9}{-2} = \frac{9}{2} = 4.5$. Not a power of 2. - $n = 1$: $\frac{-8}{-1} = 8 = 2^3$. This works: $\log_2(8) = 3$.

For $n \geq 10$: Both $n - 9$ and $n - 2$ are positive with $n - 9 < n - 2$, so $0 < \frac{n-9}{n-2} < 1$. The integer powers of 2 in $(0, 1)$ are $2^{-1} = \frac{1}{2}$, $2^{-2} = \frac{1}{4}$, $2^{-3} = \frac{1}{8}$, and so on. For larger negative exponents, $\frac{n-9}{n-2}$ becomes too small for $n \leq 20$.

Solve $\frac{n-9}{n-2} = \frac{1}{2}$: Cross-multiply to get $2(n - 9) = n - 2$, so $n = 16$. Check: $\frac{7}{14} = \frac{1}{2}$. Valid.

Solve $\frac{n-9}{n-2} = \frac{1}{4}$: $4(n - 9) = n - 2$ gives $3n = 34$, so $n = \frac{34}{3}$. Not an integer.

Solve $\frac{n-9}{n-2} = \frac{1}{8}$: $8(n - 9) = n - 2$ gives $7n = 70$, so $n = 10$. Check: $\frac{1}{8}$. Valid.

Solve $\frac{n-9}{n-2} = \frac{1}{16}$: $16(n - 9) = n - 2$ gives $15n = 142$, so $n = \frac{142}{15}$. Not an integer.

Smaller fractions push n below 10, so no further solutions exist.

Conclusion. The valid integers are $n = 1$, $n = 10$, and $n = 16$. There are 3 such values.

Question 9

Answer: C

The curve C has equation $y = x^3 - 6x^2 + 4x$. The tangent to C at the point where $x = 1$ intersects C again at the point P . The tangent to C at P intersects C again at the point Q .

Determine the x -coordinate of Q .

Use Vieta's formulas for cubics. When a line intersects $y = x^3 + bx^2 + \dots$, the sum of the three intersection x -coordinates equals $-b$ (the coefficient of x^3 is 1, and the line does not alter the x^3 or x^2 terms).

For $y = x^3 - 6x^2 + 4x$, the sum of roots is always $-(-6) = 6$.

From $x = 1$ to P : A tangent at $x = 1$ counts as a double root, so the three roots are $1, 1, x_P$:

$$1 + 1 + x_P = 6 \implies x_P = 4$$

From P to Q : The tangent at $x = 4$ gives roots $4, 4, x_Q$:

$$4 + 4 + x_Q = 6 \implies x_Q = -2$$

(An interactive version of this diagram is on tmua.co.uk.)

The x -coordinate of Q is **-2**.

Question 10

Answer: B

What is the shortest distance between the graphs of the following two equations in the xy -plane?

$$|x| + |y| = 6$$

$$|x - 1| + |y - 4| = 13$$

Identify the shapes. The equation $|x| + |y| = r$ describes a **diamond** (square rotated 45°) with vertices at $(\pm r, 0)$ and $(0, \pm r)$, whose sides have equations $x \pm y = \pm r$.

Diamond 1: $|x| + |y| = 6$, centred at the origin, with sides $x + y = 6$, $x - y = 6$, $x + y = -6$, $x - y = -6$.

Diamond 2: $|x - 1| + |y - 4| = 13$, centred at $(1, 4)$, with sides $x + y = 18$, $x - y = -16$, $x + y = -8$, $x - y = 10$.

(An interactive version of this diagram is on tmua.co.uk.)

Find parallel side distances. The shortest distance between two non-intersecting shapes with parallel sides is the perpendicular distance between the closest pair of parallel sides. For lines $Ax + By = C_1$ and $Ax + By = C_2$, the distance is $\frac{|C_1 - C_2|}{\sqrt{A^2 + B^2}}$.

Compare all four pairs:

Direction	Diamond 1 side	Diamond 2 side	Distance
$x + y$	$x + y = -6$	$x + y = -8$	$\frac{ -6 - (-8) }{\sqrt{2}} = \frac{2}{\sqrt{2}} = \sqrt{2}$
$x - y$	$x - y = 6$	$x - y = 10$	$\frac{ 10 - 6 }{\sqrt{2}} = 2\sqrt{2}$
$x - y$	$x - y = -6$	$x - y = -16$	$\frac{10}{\sqrt{2}} = 5\sqrt{2}$
$x + y$	$x + y = 6$	$x + y = 18$	$\frac{12}{\sqrt{2}} = 6\sqrt{2}$

The minimum is $\sqrt{2}$, occurring between the sides $x + y = -6$ and $x + y = -8$.

The answer is $\sqrt{2}$. A rough sketch is fully sufficient here. Should have done an art degree.

Question 11

Answer: E

The sum S_n is defined by

$$S_n = \sum_{r=1}^n \log_2 \left(1 + \frac{2r+3}{r(r+2)} \right)$$

Find the smallest integer n such that $S_n \geq 4 + \log_2 5$.

Simplify the argument of the logarithm. Combine the 1 with the fraction:

$$1 + \frac{2r+3}{r(r+2)} = \frac{r(r+2) + 2r+3}{r(r+2)} = \frac{r^2 + 4r + 3}{r(r+2)} = \frac{(r+1)(r+3)}{r(r+2)}$$

Each term of S_n becomes $\log_2 \left(\frac{(r+1)(r+3)}{r(r+2)} \right)$, which splits as:

$$\log_2 \left(\frac{r+1}{r} \right) + \log_2 \left(\frac{r+3}{r+2} \right)$$

Evaluate both telescoping sums. Summing from $r = 1$ to n :

First sum: $\sum_{r=1}^n (\log_2(r+1) - \log_2 r)$ telescopes to $\log_2(n+1) - \log_2 1 = \log_2(n+1)$.

Second sum: $\sum_{r=1}^n (\log_2(r+3) - \log_2(r+2))$ telescopes to $\log_2(n+3) - \log_2 3$.

Combining both:

$$S_n = \log_2 \left(\frac{(n+1)(n+3)}{3} \right)$$

Set up the inequality. Since $4 + \log_2 5 = \log_2 16 + \log_2 5 = \log_2 80$:

$$\log_2 \left(\frac{(n+1)(n+3)}{3} \right) \geq \log_2 80$$

Since $\log_2$ is increasing, this reduces to:

$$\frac{(n+1)(n+3)}{3} \geq 80 \implies (n+1)(n+3) \geq 240$$

Check boundary values. Solving $n^2 + 4n + 3 = 240$ gives $n \approx 13.5$, so test $n = 13$ and $n = 14$: - $n = 13$: $(14)(16) = 224 < 240$. Too small. - $n = 14$: $(15)(17) = 255 \geq 240$. Works.

(An interactive version of this diagram is on tmua.co.uk.)

The smallest integer n is **14**.

Question 12

Answer: F

Consider the equation:

$$4 \sin x \cos x + 2\sqrt{2} \cos x = k(2 \sin x + \sqrt{2})$$

Find the sum of the squares of all possible real values for the constant k such that the equation has exactly three distinct solutions for x in the range $0 \leq x < 2\pi$.

The key move is factorising. Rearrange and spot the common bracket:

$$2 \cos x(2 \sin x + \sqrt{2}) - k(2 \sin x + \sqrt{2}) = 0$$

$$(2 \sin x + \sqrt{2})(2 \cos x - k) = 0$$

Factor 1: $\sin x = -\frac{\sqrt{2}}{2}$ gives $x = \frac{5\pi}{4}$ and $x = \frac{7\pi}{4}$ – always exactly **2 roots** in $[0, 2\pi)$.

Factor 2: $\cos x = \frac{k}{2}$ – the number of roots depends on k .

For exactly **3 total** roots, Factor 2 must add exactly **1 new** root. This happens two ways:

(a) Factor 2 has exactly 1 root (not already in Factor 1). This occurs at the extreme values: $k = 2$ (giving $x = 0$) and $k = -2$ (giving $x = \pi$). Neither coincides with $\frac{5\pi}{4}$ or $\frac{7\pi}{4}$, so both work.

(b) Factor 2 has 2 roots, but one overlaps with Factor 1. If $x = \frac{5\pi}{4}$ satisfies $\cos x = \frac{k}{2}$, then $k = 2 \cos \frac{5\pi}{4} = -\sqrt{2}$. The other root of $\cos x = -\frac{\sqrt{2}}{2}$ is $x = \frac{3\pi}{4}$ (new). Similarly, overlap at $x = \frac{7\pi}{4}$ gives $k = \sqrt{2}$, with the other root $x = \frac{\pi}{4}$ (new).

(An interactive version of this diagram is on tmua.co.uk.)

The four valid values are $k = 2, -2, \sqrt{2}, -\sqrt{2}$. The sum of their squares:

$$4 + 4 + 2 + 2 = 12$$

Question 13

Answer: D

Let $f(y) = y^2 - 2$.

Find the number of distinct real solutions x to the equation $f(f(f(|x|))) = 0$

Solve $f(f(f(|x|))) = 0$ where $f(y) = y^2 - 2$ by working backwards.

Set $z_1 = |x|$ (so $z_1 \geq 0$), $z_2 = f(z_1)$, and $z_3 = f(z_2)$. The equation becomes $f(z_3) = 0$.

Solve $f(z_3) = 0$: $z_3^2 = 2$, giving $z_3 = \pm\sqrt{2}$.

Solve $f(z_2) = z_3$: $z_2^2 = 2 + z_3$.

- $z_3 = \sqrt{2}$: $z_2^2 = 2 + \sqrt{2} > 0$, giving $z_2 = \pm\sqrt{2 + \sqrt{2}}$. - $z_3 = -\sqrt{2}$: $z_2^2 = 2 - \sqrt{2} > 0$ (since $2 > \sqrt{2}$), giving $z_2 = \pm\sqrt{2 - \sqrt{2}}$.

This yields **4 distinct values** for z_2 .

Solve $f(z_1) = z_2$: $z_1^2 = 2 + z_2$. Check each case:

- $z_2 = \sqrt{2 + \sqrt{2}}$: $z_1^2 = 2 + \sqrt{2 + \sqrt{2}} > 0$. Two values. - $z_2 = -\sqrt{2 + \sqrt{2}}$: $z_1^2 = 2 - \sqrt{2 + \sqrt{2}}$.

Positive since $4 > 2 + \sqrt{2}$, i.e. $2 > \sqrt{2}$. Two values. - $z_2 = \sqrt{2 - \sqrt{2}}$: $z_1^2 = 2 + \sqrt{2 - \sqrt{2}} > 0$.

Two values. - $z_2 = -\sqrt{2 - \sqrt{2}}$: $z_1^2 = 2 - \sqrt{2 - \sqrt{2}}$. Positive since $4 > 2 - \sqrt{2}$. Two values.

All 4 cases yield 2 solutions each, giving **8 distinct values** for z_1 . These come in $\pm$ pairs: 4 positive, 4 negative.

Solve $|x| = z_1$: Since $|x| \geq 0$, only the 4 positive values of z_1 are valid. None equals zero (since every $z_1^2 > 0$). Each positive value k gives $x = k$ and $x = -k$, producing 2 solutions.

Total: $4 \times 2 = 8$ distinct real solutions.

Question 14

Answer: C

Let a and b be positive real numbers. For every real x the following inequality holds: $a \leq \frac{4}{1+e^x} + \frac{1}{1+e^{-x}} + 1 \leq b$. What is the minimum possible value of $b - a$?

Simplify the expression by substituting $t = e^x$ where $t > 0$.

The negative exponent term becomes:

$$\frac{1}{1+e^{-x}} = \frac{1}{1+1/t} = \frac{t}{t+1}$$

Combine all three parts:

$$\frac{4}{1+t} + \frac{t}{1+t} + 1 = \frac{4+t}{1+t} + 1 = \frac{5+2t}{1+t}$$

Perform polynomial division to extract the constant part:

$$\frac{5+2t}{1+t} = \frac{2(1+t)+3}{1+t} = 2 + \frac{3}{1+t}$$

Determine the range. The term $\frac{3}{1+t}$ is strictly decreasing for $t > 0$: - As $t \rightarrow 0^+$ (i.e. $x \rightarrow -\infty$): $\frac{3}{1+t} \rightarrow 3$, so the expression $\rightarrow 5$ - As $t \rightarrow \infty$ (i.e. $x \rightarrow +\infty$): $\frac{3}{1+t} \rightarrow 0$, so the expression $\rightarrow 2$

Neither bound is attained, so the range is the **open interval** $(2, 5)$.

(An interactive version of this diagram is on tmua.co.uk.)

The curve approaches but never reaches the dashed bounds at $y = 2$ and $y = 5$.

Find the minimum $b - a$. For the inequality $a \leq f(x) \leq b$ to hold for every real x , we need $a \leq 2$ (the infimum) and $b \geq 5$ (the supremum). The tightest valid choice is $a = 2$ and $b = 5$.

Note: Since the range is the open interval $(2, 5)$, the values $a = 2$ and $b = 5$ represent the tightest bounds — the function approaches but never reaches these values.

The answer is $b - a = 5 - 2 = 3$.

Question 15

Answer: D

For a real number x , let $[x]$ denote the largest integer less than or equal to x .

Find the value of

$$\int_{-4}^4 (6x^2 + [x^3]) \, dx$$

Split the integral: $I = \int_{-4}^4 6x^2 \, dx + \int_{-4}^4 [x^3] \, dx$.

Part 1: Since $6x^2$ is an even function:

$$\int_{-4}^4 6x^2 \, dx = 2 \int_0^4 6x^2 \, dx = 2 \left[2x^3 \right]_0^4 = 2(128) = 256$$

Part 2: Use the symmetry trick. For any function on a symmetric interval:

$$\int_{-4}^4 [x^3] \, dx = \int_0^4 ([x^3] + [-x^3]) \, dx$$

The **key floor function property** states: for any real y ,

$$[y] + [-y] = \begin{cases} 0 & \text{if } y \in \mathbb{Z} \\ -1 & \text{if } y \notin \mathbb{Z} \end{cases}$$

Since x^3 is an integer only at isolated points (a set of measure zero, which does not affect the integral), $[x^3] + [-x^3] = -1$ almost everywhere on $[0, 4]$:

$$\int_0^4 (-1) \, dx = -4$$

Combining: $I = 256 + (-4) = \mathbf{252}$. *This one is gruelling. Don't you let it go.*

Question 16

Answer: A

Evaluate the sum

$$\frac{1}{\log_2(64!)} + \frac{1}{\log_3(64!)} + \frac{1}{\log_4(64!)} + \cdots + \frac{1}{\log_{64}(64!)}.$$

Apply the change-of-base identity. For any base k , the reciprocal of $\log_k(64!)$ can be rewritten using $\frac{1}{\log_k N} = \log_N k$. This follows directly from the change-of-base formula: $\log_k N = \frac{\ln N}{\ln k}$, so $\frac{1}{\log_k N} = \frac{\ln k}{\ln N} = \log_N k$.

Rewrite every term. Each summand becomes:

$$\frac{1}{\log_k(64!)} = \log_{64!}(k)$$

Combine the sum using logarithm properties. The full sum is:

$$\sum_{k=2}^{64} \log_{64!}(k) = \log_{64!}(2) + \log_{64!}(3) + \cdots + \log_{64!}(64)$$

Since $\log_b(m) + \log_b(n) = \log_b(mn)$, this telescopes into:

$$\log_{64!}(2 \times 3 \times 4 \times \cdots \times 64) = \log_{64!}(64!)$$

Evaluate. Since $\log_{64!}(64!) = 1$, the value of the sum is **1**. *The change-of-base step above is of course an application of the Taylor Swift change-of-base law.*

Question 17

Answer: A

Let

$$S = \sum_{r=0}^{19} (-1)^r \binom{32}{r} \binom{32}{13+r}.$$

Evaluate S .

Apply the symmetry identity $\binom{n}{k} = \binom{n}{n-k}$ to the second binomial coefficient:

$$\binom{32}{13+r} = \binom{32}{19-r}$$

This transforms the sum into:

$$S = \sum_{r=0}^{19} (-1)^r \binom{32}{r} \binom{32}{19-r}$$

Recognise this as a coefficient extraction. The product $(1+x)^{32}(1-x)^{32}$ expands via the binomial theorem as:

$$\sum_j \sum_k \binom{32}{j} \binom{32}{k} (-1)^k x^{j+k}$$

The coefficient of x^{19} collects all pairs where $j+k=19$, i.e. $j=19-k$:

$$[x^{19}] = \sum_{k=0}^{19} \binom{32}{19-k} \binom{32}{k} (-1)^k$$

This is exactly our sum S . Therefore $S = [x^{19}]((1+x)^{32}(1-x)^{32})$.

Simplify the generating function. Combine the product:

$$(1+x)^{32}(1-x)^{32} = (1-x^2)^{32}$$

The expansion of $(1-x^2)^{32}$ contains only **even powers** of x : the terms are $x^0, x^2, x^4, \dots, x^{64}$. Since 19 is odd, the coefficient of x^{19} is **0**.

The answer is $S = 0$.

Question 18

Answer: D

Let θ be a real number. Define two infinite series:

$$S_1(\theta) = \sin^2 \theta + \sin^4 \theta + \sin^6 \theta + \dots$$

$$S_2(\theta) = \cos^2 \theta + \cos^4 \theta + \cos^6 \theta + \dots$$

Both series converge whenever $|\sin \theta| < 1$ and $|\cos \theta| < 1$.

Suppose that

$$S_1(\theta) = \frac{2}{3} \quad \text{and} \quad S_2(\theta) = \frac{3}{2}.$$

How many distinct real values of θ are there in the range $-2\pi \leq \theta \leq 2\pi$ which satisfy both equations?

Recognise geometric series. Each series has first term a and common ratio r equal to $\sin^2 \theta$ and $\cos^2 \theta$ respectively:

$$S_1 = \frac{\sin^2 \theta}{1 - \sin^2 \theta}, \quad S_2 = \frac{\cos^2 \theta}{1 - \cos^2 \theta}$$

Solve $S_1 = \frac{2}{3}$:

$$\frac{\sin^2 \theta}{1 - \sin^2 \theta} = \frac{2}{3} \implies 3 \sin^2 \theta = 2 - 2 \sin^2 \theta \implies \sin^2 \theta = \frac{2}{5}$$

Verify $S_2 = \frac{3}{2}$: From the Pythagorean identity, $\cos^2 \theta = 1 - \frac{2}{5} = \frac{3}{5}$. Then:

$$S_2 = \frac{3/5}{1 - 3/5} = \frac{3/5}{2/5} = \frac{3}{2} \quad \checkmark$$

So the second equation is automatically satisfied. The problem reduces to counting solutions of $\sin^2 \theta = \frac{2}{5}$ in $[-2\pi, 2\pi]$.

Count solutions. The function $\sin^2 \theta$ has period π (since $\sin^2(\theta + \pi) = \sin^2 \theta$). In each interval of length π , there are exactly 2 solutions to $\sin^2 \theta = \frac{2}{5}$ (one from $\sin \theta = +\sqrt{2/5}$ and one from $\sin \theta = -\sqrt{2/5}$).

The interval $[-2\pi, 2\pi]$ has length 4π , containing 4 full periods.

Total solutions: $2 \times 4 = 8$.

Question 19

Answer: D

Find the sum of all distinct real solutions to the following equation:

$$(x^2 - 7x + 11)^{x^2 - 2x - 2} = 1$$

For $(f(x))^{g(x)} = 1$, where $f(x) = x^2 - 7x + 11$ and $g(x) = x^2 - 2x - 2$, there are three cases:

Case 1: Base equals 1. $x^2 - 7x + 11 = 1 \implies x^2 - 7x + 10 = 0 \implies (x - 2)(x - 5) = 0$.
Solutions: $x = 2, 5$.

Case 2: Base equals -1 and exponent is even. $x^2 - 7x + 11 = -1 \implies x^2 - 7x + 12 = 0 \implies (x - 3)(x - 4) = 0$.

Check exponents: - $x = 3$: $g(3) = 9 - 6 - 2 = 1$ (odd). $(-1)^1 = -1 \neq 1$. **Not valid.** - $x = 4$: $g(4) = 16 - 8 - 2 = 6$ (even). $(-1)^6 = 1$. **Valid.**

Case 3: Exponent equals 0 and base is non-zero. $x^2 - 2x - 2 = 0 \implies x = 1 \pm \sqrt{3}$.

Check bases are non-zero: - $x = 1 + \sqrt{3}$: $f(1 + \sqrt{3}) = (1 + \sqrt{3})^2 - 7(1 + \sqrt{3}) + 11 = 1 + 2\sqrt{3} + 3 - 7 - 7\sqrt{3} + 11 = 8 - 5\sqrt{3} \neq 0$. **Valid.** - $x = 1 - \sqrt{3}$: $f(1 - \sqrt{3}) = (1 - \sqrt{3})^2 - 7(1 - \sqrt{3}) + 11 = 1 - 2\sqrt{3} + 3 - 7 + 7\sqrt{3} + 11 = 8 + 5\sqrt{3} \neq 0$. **Valid.**

All five solutions: $x = 2, 5, 4, 1 + \sqrt{3}, 1 - \sqrt{3}$.

Sum: $2 + 5 + 4 + (1 + \sqrt{3}) + (1 - \sqrt{3}) = 2 + 5 + 4 + 1 + 1 = 13$.

The answer is **13**. *Three cases, one parity check, and we're good to go.*

Question 20

Answer: F

Let the “bimode” of a data set be equal to the regular mode if there is a singular most common value, but equal to the mean of the modes if there is no unique modal value.

What is the sum of all real values of x for which the mean, median and bimode (in any order) of the seven numbers

$$-2, 1, 1, 9, 9, 12, x$$

form an arithmetic sequence?

Compute each statistic in terms of x .

The sum of the six known values is $-2 + 1 + 1 + 9 + 9 + 12 = 30$, so the **mean** is $\frac{30+x}{7}$.

The **median** is the 4th value once all seven numbers are sorted. Depending on where x falls:

- $x \leq 1$: median = 1 - $1 < x < 9$: median = x - $x \geq 9$: median = 9

The **bimode**: since 1 and 9 each appear twice, they tie for mode, giving bimode = $\frac{1+9}{2} = 5$. However, if $x = 1$ or $x = 9$, one value appears three times, making it the unique mode (bimode = 1 or 9 respectively).

Set up the arithmetic sequence condition. Three numbers a, b, c form an arithmetic sequence when one of them equals the average of the other two. We check each interval for x .

Case A: $x \leq 1$ (excluding $x = 1$). Median = 1, bimode = 5, mean = $\frac{30+x}{7}$. The three values $\{\frac{30+x}{7}, 1, 5\}$ form an AP when: - Mean = 3 (the average of 1 and 5): $\frac{30+x}{7} = 3 \Rightarrow x = -9$. Valid ($-9 \leq 1$). - 1 is the average of mean and 5: mean = -3 , so $x = -51$. Valid. - 5 is the average of mean and 1: mean = 9, so $x = 33$. Not in this interval.

Solutions: $x = -9, -51$.

Case B: $1 < x < 9$. Median = x , bimode = 5. For x to be the middle term of the AP, $x = \frac{1}{2}\left(\frac{30+x}{7} + 5\right)$, which simplifies to $x = 5$. Checking the other orderings yields no additional solutions in this interval.

Solution: $x = 5$.

Case C: $x > 9$ (excluding $x = 12$). Median = 9, bimode = 5, mean = $\frac{30+x}{7}$. The three values $\{\frac{30+x}{7}, 9, 5\}$ form an AP when: - Mean = 7 (average of 5 and 9): $x = 19$. Valid. - Mean = 13 (so 9 is the average of 5 and 13): $x = 61$. Valid.

Solutions: $x = 19, 61$.

Case D: Special values $x = 1, 9, -2, 12$. Quick checks show none produce a valid arithmetic sequence.

Sum all solutions. The five valid values are $-51, -9, 5, 19, 61$:

$$-51 + (-9) + 5 + 19 + 61 = \mathbf{25}$$

The bimode is not a real statistical object. It should be.